MD ISMAIL HOSSAIN

Graduate Teaching Assistant | School of Computing | University of Wyoming, Laramie, USA

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EDUCATION

M.S. in Geospatial Information Science & Technology *Fall 2025 – Present*
School of Computing, University of Wyoming, USA · GPA: 4.00/4.00 *(Expected Spring, 2027)*
Thesis: Digital Twin Readiness and Data Equality in Small Cities (ongoing)
Chapter 1: When Urban Digital Twins Reproduce or Reduce Inequality: A Twin-Pole Framework Across the Digital Twin Lifecycle
Chapter 2: Governing and Implementing the Small-City Digital Twin: Data Privacy, Sharing, and Equity in Laramie and Afton, Wyoming.
Supervisor: Dr. Jason "Jake" Hawes

B.S. in Local Government & Urban Development
Jatiya Kabi Kazi Nazrul Islam University, Bangladesh · CGPA: 3.84/4.00 · Ranked 1st in class of 55 *2020 – 2025*
Thesis: Understanding Teleworking Adoption and Its Impact on Transportation in Bangladesh: A PLS-SEM Analysis

RESEARCH INTERESTS

- Travel Behavior
- Sustainable Transportation Planning
- Digital Twins
- Spatial Data Science
- Traffic Safety
- Land Use & Transportation
- GeoAI & ML
- Human-Environment Systems

RESEARCH, PUBLICATIONS & PROJECTS

(* = Corresponding Author)

Peer-Reviewed Journal Articles

1. **Hossain, M. I.,** Mazhar, M., & Haque, M. M., * Rahman, M.M., & Rahman, A. (2026). Service Quality, User Heterogeneity, and Policy Priorities in Three Wheeler Paratransit: A Suburban Case Study from Trishal, Bangladesh. *Case Studies on Transport Policy*. https://doi.org/10.1016/j.cstp.2026.101998

2. * **Hossain, M. I.,** Lipi, A. I., & Sultana, P. (2026). Local perceptions of climate change impacts and adaptation strategies for rural infrastructure in the Chalan Beel area of Bangladesh. *Journal of Critical Infrastructure Policy, 7*(2), e70037. https://doi.org/10.1002/jci3.70037

3. * Hasan, S. H., **Hossain, M. I.,** et al. Mobile Banking Usage for Secondary School Stipends in Bangladesh: A Cross-Sectional Survey of Students in 2023. *American Journal of Social Development and Entrepreneurship, 3(1).* https://doi.org/10.54536/ajsde.v3i1.7018

Peer-Reviewed Conference Proceedings

1. * Njoku, J. N., Shukla, D., Katariya, V., **Hossain, M. I.**, & Hawes, J. K. (in presss) PANDA: A Lightweight Digital Twin Framework for Smart Parking Management. Proceedings of the *i3CE Conference 2026, Songdo, South Korea*. PDF File (Forthcoming in ASCE Conference Proceedings.)

Manuscripts Under Review

1. * **Hossain, M. I.,** Rahman, M. T., Heatwave-adaptive travel behavior in Bangladesh: a four-channel analysis of mobility reduction, mode choice, temporal, and digital substitution. *Transportation* (under review). SSRN Preprint: https://dx.doi.org/10.2139/ssrn.5056614

2. **Hossain, M. I.,** * Hawes, J. K., Lueck, V., Njoku, J. N., Hamerlinck, J., & Katariya, V. (2026). (In preparation) When Urban Digital Twins Reproduce or Reduce Inequality: A Twin-Pole Framework Across the Digital Twin Lifecycle. *Target Journal: Cities*.

3. * **Hossain, M. I.,** Sayem, M. A., Rabbani, M. G., & Makhdum, N. (2025) Willing but unable? A willingness–ability framework for teleworking adoption, workplace access, and perceived transport impacts in Bangladesh. *Transportation Planning and Technology* (under review).

4. Rahman, M. M., Begum, H., & * **Hossain, M. I.** (2026). Beyond the thirty-percent rule: A machine-learning triangulation of affordability and habitability in Dhaka. *Journal of Housing and Built Environment* (Under Review)

5. * **Hossain, M. I.,** Ahmed, R., & Sayem, M. A. (2026). The leapfrog gap as a smart-city planning diagnostic: mobile money demand and cash-only fare collection at Dhaka Metrorail. *Cities: Smarter Future* (Under review).

6. * **Hossain, M. I.,** Shanta, F., Rahman, M. M., Haque, M. M. (2026). Traffic congestion as a barrier to health access and spatial justice in a secondary city hospital corridor in Mymensingh, Bangladesh. *Frontiers of Urban and Rural Planning* (Under review)

7. **Hossain, M. I.,** Khan, S., * Ksaibati, K. (2026). Ice with and without a weather cue: speed adaptation, injury severity, and spatial priorities in Wyoming winter crashes. *IATSS Research* (Submitted)

8. * **Hossain, M. I.,** Parthib, M.S.K., Ahmed, Z.B. (2026) Does urbanization equalize? Re-examining the urban Kuznets curve in Bangladesh, 1991–2022, with survey-year evidence on income and consumption inequality. *Discover Cities* (Submitted)

9. * **Hossain, M. I.,** Ahmed, R., Mostofa, S., Islam, M.M. (2026). Peri-urban land transformation at the edge of a South Asian megacity: GIS-based evidence from Keraniganj, Dhaka, 2015–2025. *Frontiers of Urban and Rural Planning (Under Review)*

Published Book Chapter *2022*

1. **Hossain, M. I.** (2022). *Jatiya Kabi Kazi Nazrul Islam University, Bangladesh.*

Research Funding *2024*

1. "Urban Society and Political Reflections in Nazrul's (National Poet of Bangladesh) Plays" (50,000 BDT Grant, 2024), A prestigious fully funded research grant by Institute of Nazrul Studies, Jatiya Kabi Kazi Nazrul Islam University.

Invited Talks

1. **Hossain, M. I.** (2026, February 23). *Digital Twin Readiness and Data Equality in Small Cities* [Lightning talk]. Digital Twin Community of Practice Organizational Event, School of Computing, University of Wyoming, Laramie, WY.

Conference and Poster Presentations

1. **Hossain, M. I.,** Ullah, I., Khan, S., & Ksaibati, K. (2026). Winter-related crashes in Wyoming: Crash speed, injury severity, and crash hotspots. Poster presentation, Southern Colorado Transportation Conference 2026.
2. **Hossain, M. I.**, Hawes, J. K., & Hamerlinck, J. D. 'Digital Twin Readiness and Data Equality in Small Cities: A Comparative Study between Laramie and Afton, Wyoming.' (Abstract) *AAG Annual Meeting, San Francisco, CA, 2026.*
3. **Hossain, M. I.**, Hawes, J. K., & Hamerlinck, J. D. Poster presented at the *UW Science Institute Annual Spring Research Symposium, University of Wyoming, Laramie, WY. Spring 2026.*

WYDOT School Safety Project — Lead Analyst *Fall 2025*

GIST 5200: Geographic Visualization, University of Wyoming

- Led GIS-based analysis with Wyoming DOT to assess school-zone safety across the state.
- Built spatial visualizations in ArcGIS Pro to identify high-risk traffic and safety patterns.
- Coordinated team workflow and all presentation outputs as project lead.

TECHNICAL SKILLS

GIS & Remote Sensing	ArcGIS Pro · QGIS · Geodata · Google Earth Engine · Spatial Analysis
Programming	Python · R · Java · JavaScript · C#
Machine Learning	Predictive Analytics · Data Modeling · Regression
Data Management	MySQL · SQLite · KoBo Toolbox (survey & field data)
Statistical Analysis	SPSS · Jamovi · Parametric & Non-parametric Tests · Regression Modeling · SmartPLS
Dev Tools	VS Code · RStudio · Anaconda · GitHub · GitLab

AWARDS & SCHOLARSHIPS

- CaGIS Master's Scholarship Award, Cartography and Geographic Information Society *2026*
- NSF Fully Funded AI-Unlocked/ACCESS Workshop, Denver, Colorado *2026*
- Graduate Student Travel Funding Scholarship, University of Wyoming *Spring 2026*
- Vice Chancellor Merit Scholarship (Top-ranked student in dept, 4×) *2021–2024*
- Departmental Merit Scholarship (Top-ranked student, 3×) *2023–2025*
- 1st Place — University & Tarash Student Assoc. Essay Competitions *2021*

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant *Fall 2026 – Present*

School of Computing, University of Wyoming, Laramie, USA

- Support undergraduate GIST 2310: Introduction to GIS laboratory instruction and hands-on geospatial exercises.
- Assist students with GIS concepts, spatial data, mapping, assignments, and spatial analysis techniques.

Graduate Research Assistant *Fall 2025 – Summer 2026*

School of Computing, University of Wyoming, Laramie, USA

Smart Rural Places Project: Digital Twins for Rural Places | *Project Website:* ***https://datahive.uwyo.edu/rural-smart-digital-twin/***

- Conducting thesis research on digital twin readiness and geospatial data equality, comparing a university-town (Laramie) and a rural community (Afton) in Wyoming.
- Geospatial data collection, spatial analysis using GIS and programming tools.
- Support interdisciplinary research on urban systems, infrastructure, and academic writing.

Undergraduate Research Assistant *2022 – 2023*

Dept. of Local Government & Urban Development, JKKNIU, Bangladesh

- Literature reviews on climate resilience; field data collection via surveys, interviews, and focus groups.

Undergraduate Teaching Assistant *2020 – 2022*

Dept. of Local Government & Urban Development, JKKNIU, Bangladesh

- Supported course preparation and student mentoring.
- Created an academic resource blog: lgudzone.blogspot.com

Lead SDET (Remote) — EagleMD Psychiatry MH Services LLC, Georgia, USA *2024 – 2025*

- Improved bug detection by 20% via optimized testing strategies; mentored junior testers.

Software Development Engineer in Test (SDET) Remote— ICTER Guru LLC, Atlanta, USA *2022 – 2024*

- Developed 50+ automated test scripts using Selenium and Java, improving testing efficiency and coverage.
- Helped junior testing engineer through online training (Class samples are in both Bengali and English: https://www.youtube.com/playlist?list=PLnPs2YiQB-Qrw5MrZuqJYBJvCV3MZNY1D)

Math & Computer Science Instructor (Remote) — ICTER Guru LLC, Atlanta, USA *2020 – 2023*

- Taught mathematics, programming, statistics & data science to high school and college students.

ENTREPRENEURSHIP & START-UP

Co-Founder & CEO — NextHunger IT Firm *2023 – Present*

https://nexthunger.vercel.app/

- Founded a tech firm specializing in custom websites, web applications, and full branding (logos, business cards, letterheads).
- Delivered 5 personal websites, 2 institutional websites, and 1 psychiatry platform with complete brand identity packages.
- Managed client relationships, project timelines, and a cross-functional development team end-to-end.

LEADERSHIP & SERVICE

Secretary — Environment, Climate, Disasters & Planning *2022 – 2025*

Tarash Student Association, Bangladesh

- Planned and coordinated academic and volunteer events on environment and community development.

Elected Class Representative (3×) *2020 – 2022*

Jatiya Kabi Kazi Nazrul Islam University, Bangladesh

- Represented students, facilitated faculty communication, and organized academic events.

Member, American Association of Geographers (AAG) *2025 – Present*

Member, American Society for Photogrammetry and Remote Sensing *2025 – Present*

Student Chapter, University of Wyoming, Laramie, USA

Sports Analyst & Content Creator | YouTube — 2022 - 2023

Ismail Times | https://www.youtube.com/@ismailtimes

• Produced data-informed analyses of cricket and soccer, interpreting player performance, match statistics, and tactical trends for online audiences.